

Abstract geometric lines in the top left corner, consisting of several thin, light brown lines that intersect and form various polygons and shapes.

INDUSTRIAL IMAGE ANOMALY DETECTION

Presenter:

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CONTEXT & MOTIVATION

Goal: Detect defect in manufacturing (scratches, cracks, contaminations)

Importance: Automative quality control and reduce inspection errors



Defects may be small or hard to detect, manual inspection can lead to errors and lead to high human cost



Need scalable, interpretable and reliable AI methods



KEY CHALLENGES

Data Imbalance: few anomaly sample

Variability of defects(shapes, texture, severity)

Real-time constraints -> solutions must be efficient

Interpretability -> operator needs clear explainability



DATASET: OVERVIEW

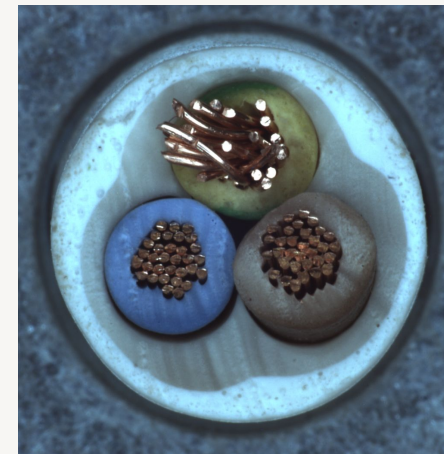
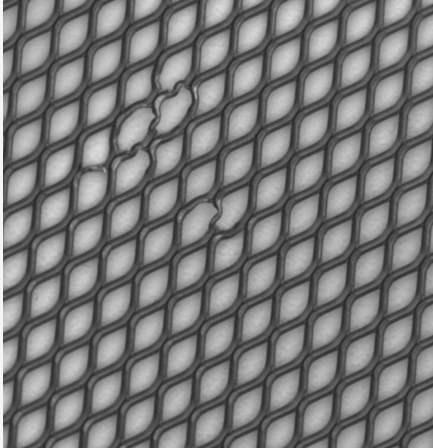
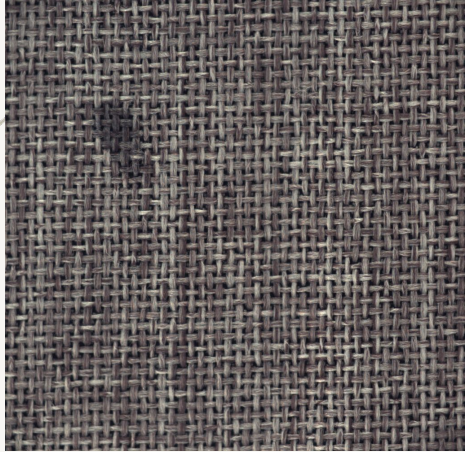
MVTec AD-V1

- A benchmark anomaly detection dataset with **over 5,000 high-resolution images** spanning **15 object** and **5 texture categories**. It includes pixel-precise anomaly annotations and separate sets of defect-free training and mixed test images.
- Ideal for initial benchmarking of unsupervised, one-class learning methods.

MVTec AD-V2

- Introduces **eight new challenging industrial inspection scenarios**, totaling **over 8,000 high-resolution images**.
- Includes multiple lighting conditions for robustness testing under **distribution shifts**.

DATASET: VISUAL EXAMPLES



METHODS LANDSCAPE

Reconstruction based (GANs, Autoencoders):
simple, but weak on subtle anomalies

Feature-based: (Patchcore, ViT): exploit
pretraining and more robust

Classification-based: not feasible in this setting
with few anomalies

RECONSTRUCTION VS FEATURE-BASED METHODS

- Image space problems:
 - In image space no semantic meaning – scratch and texture variation might look equal at pixel level
 - Sensitive to noise and light variations
- Feature space advantages:
 - Pre-trained deep networks can produce **very discriminative features**
 - Deep Features from multi-scale regional feature generator – describe both local characteristics and spatial context information

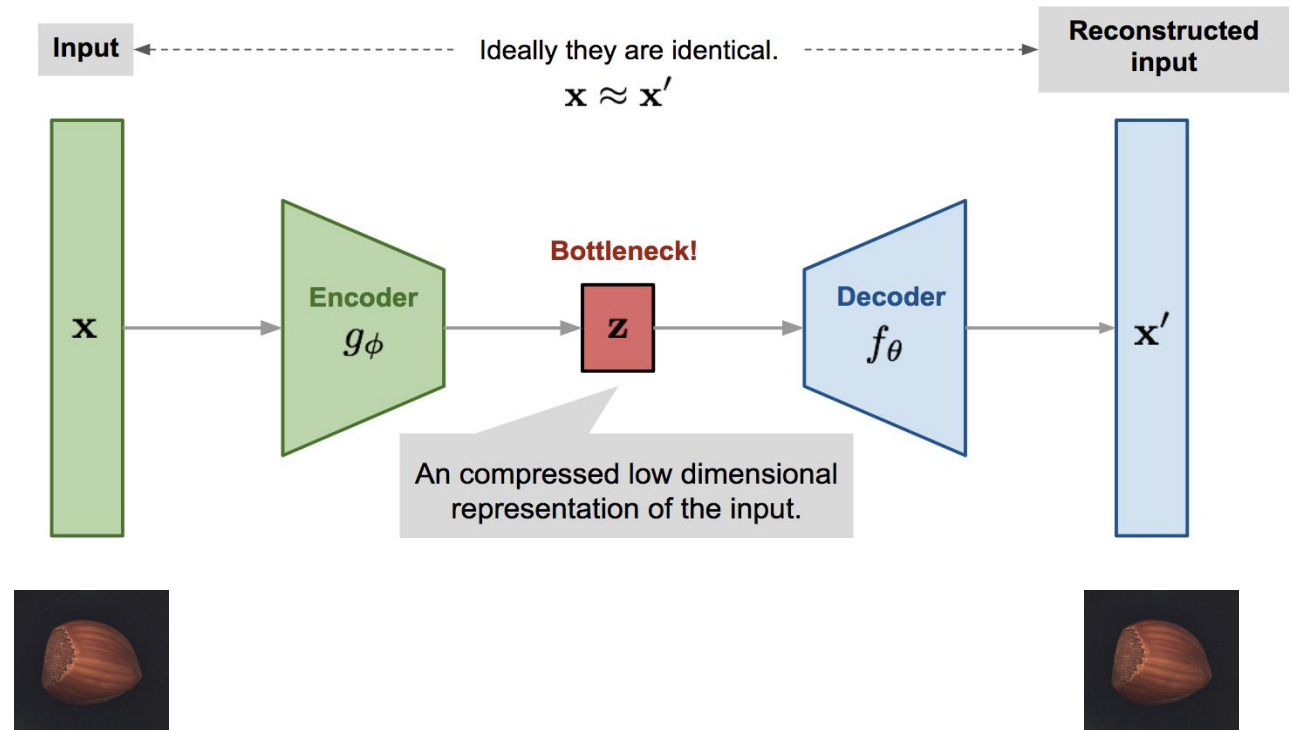
BASELINE AE: ARCHITECTURE

Encoder (Downsampling):

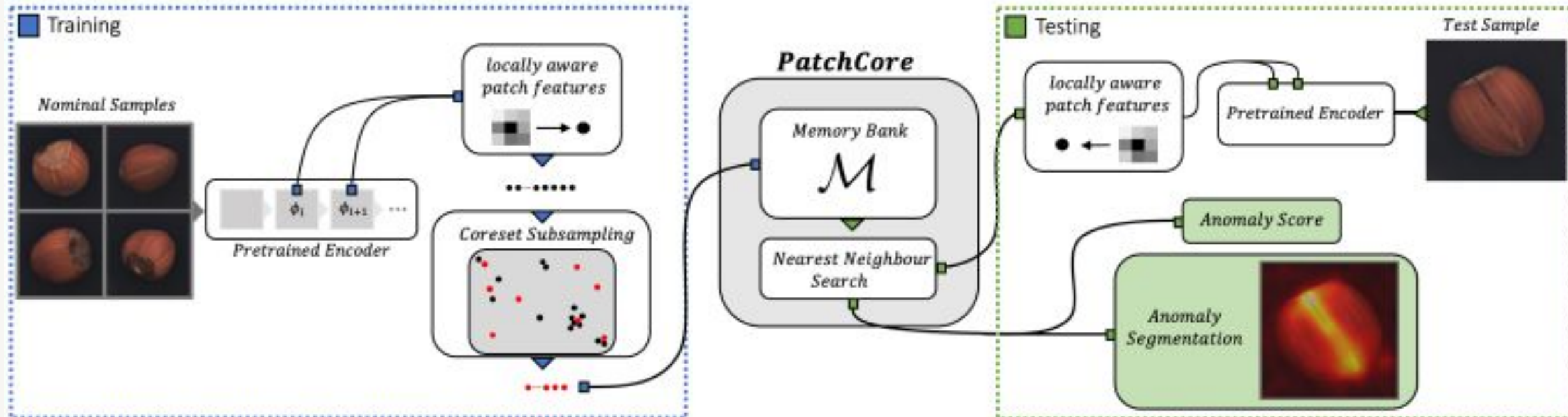
- 3 convolutional layers with progressively more channels (3→64→128→256)
- Each layer uses 3x3 kernels with stride=2 for downsampling
- ReLU activation between layers
- Reduces input from (3, 224, 224) to (256, 28, 28)

Decoder (Upsampling):

- 3 transposed convolutional layers that mirror the encoder
- Progressively fewer channels (256→128→64→3)
- Uses stride=2 and output_padding=1 for upsampling
- Final sigmoid activation to normalize output to [0,1]
- Reconstructs back to (3, 224, 224)



PATCHCORE: HOW IT WORKS



Step 1: Extract feature with a pretrained CNN backbone, in this case ResNet was used

Step 2: Build a memory bank for normal patch embeddings

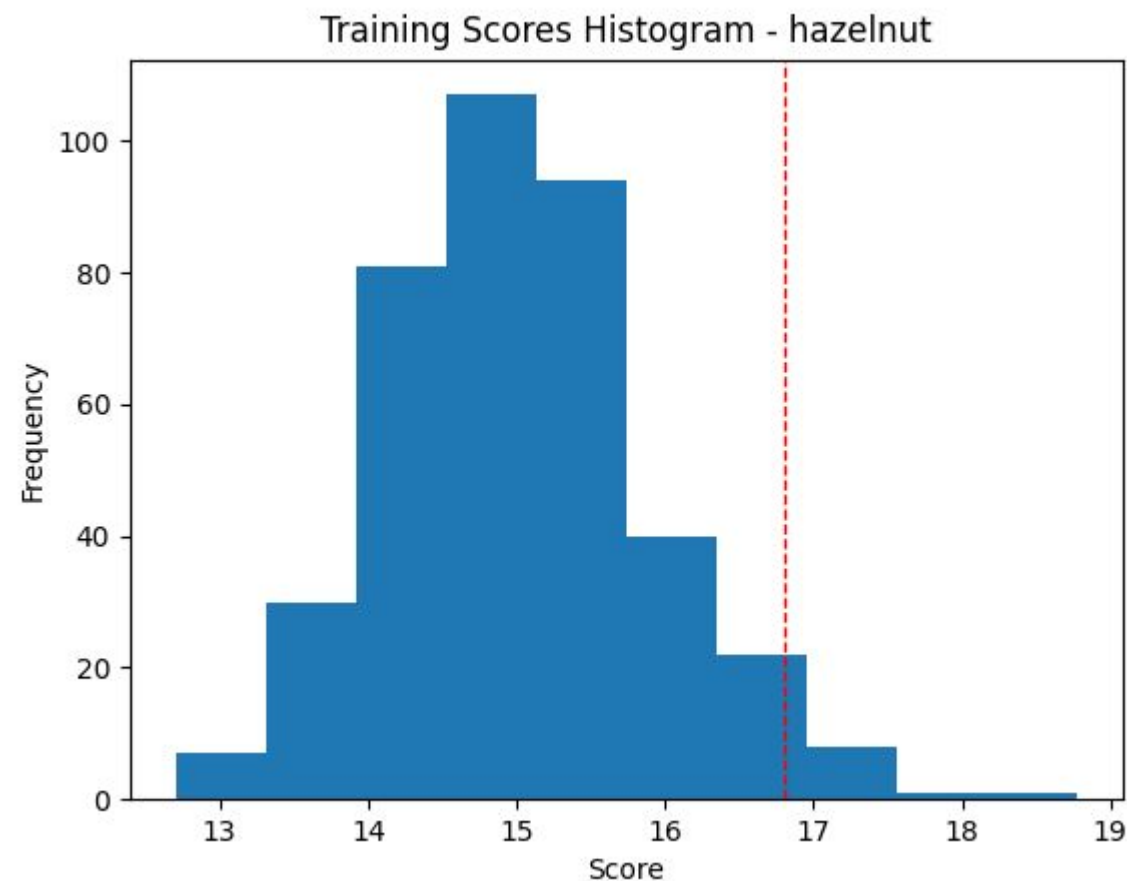
Step 3: For a test image, compare patch embeddings with memory (K-NN is used)

Step 4: Distance between closest patch in the memory bank is taken as anomaly score

PATCHCORE: THRESHOLD EXAMPLES

To decide if a patch anomalous, we assume a Gaussian distribution on the errors in the training set and we choose 2 standard deviation away from the mean

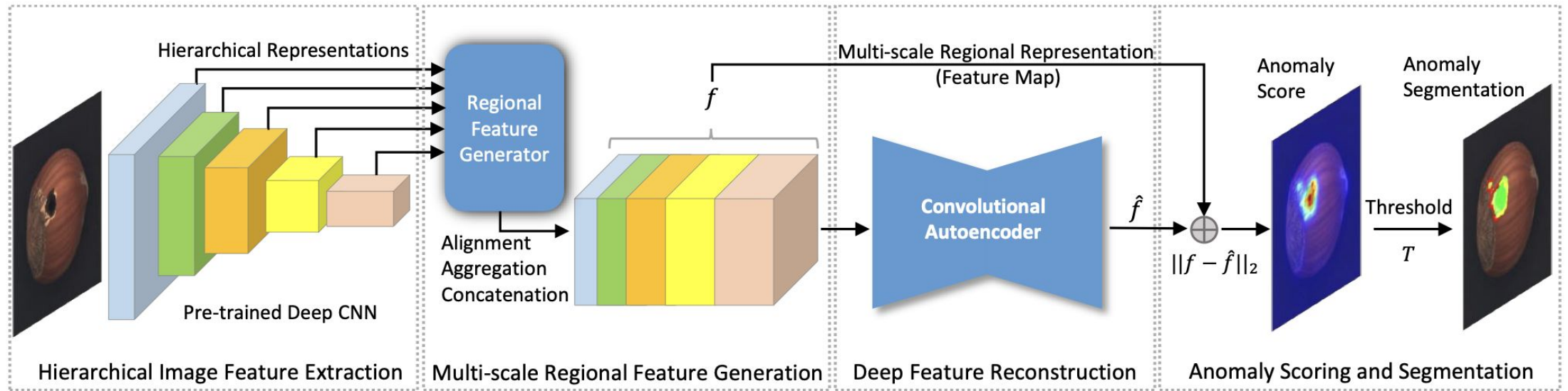
From the image we can see that the gaussian distribution is not really met but still not far away and works well in practice



DFR: PIPELINE

- Extract multi-scale regional features from a ResNet50
- Train convolutional autoencoder to reconstruct Deep Feature Maps
- Calculate feature **reconstruction error** of the Convolutional Autoencoder.
- Predict anomalies thresholding the reconstruction error map.
- Threshold calculation:
 - Adding 1 std to mean error (aggressive)
 - Adding 3 std to mean error (statistically captures about 99.7% of the data of a normal distribution)
 - Adding 5 std to mean error (very conservative)

DFR: ARCHITECTURE



- ResNet50 inherently builds hierarchical representations, concatenating features from different layers.
 - Layer 2 output shape: [512, 28, 28], stride = 8 relative to input. More local details.
 - Layer 3 output shape: [1024, 14, 14], stride = 16 relative to input. More global semantics and context.
 - Feature maps aligned, via bilinear up-sampling, and concatenated. Output shape: [1536, 28, 28]
- ResNet50 has already learned spatial patterns: autoencoder only needs to remap channel representations with **1x1 convs**.
 - Encoder: 1536 → 818 → 200 → 100 channels
 - Decoder: 100 → 200 → 818 → 1536 channels

RESULTS: MODEL COMPARISON (ROC-AUC)

	HazelNuts	Leather	Wood	Carpet	Bottle
Base	0.66	0.41	0.59	0.46	0.63
ViT	0.87	0.75	0.57	0.52	0.71
PatchCore	0.99	1.0	0.9821	0.9683	0.9975
Deep Feature Reconstruction	1.0	1.0	0.9894	0.9747	0.9492



CODE REVIEW

A series of thin, light brown lines forming an abstract geometric pattern on the left side of the slide. The lines intersect to create various polygonal shapes, some of which are nested within others, creating a sense of depth and complexity.

THANKS FOR YOUR
ATTENTION